

TUE MINH CAO

PhD Student, University of Florida · Mechanistic Interpretability & AI Safety Research

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GPA 4.0/4.0 (UF)

RESEARCH STATEMENT

My research develops methods to understand the internal mechanisms of large language models, with the goal of making AI systems more transparent, predictable, and safe. I focus on two complementary views of model internals: the activation space - via sparse autoencoders, circuit tracing, feature geometry; and the weight space — via weight decomposition and weight-level steering. My current work studies how features are structured across layers, how they interact with model weights, and how that interaction gives rise to emergent capabilities and failure modes. I believe that a unified understanding of activation and weight spaces is a necessary foundation for mechanistic alignment.

EDUCATION

PhD, Computer Science – University of Florida (UF)

Jan 2026 – Present

Advisor: Prof. My T. Thai · GPA: 4.0 / 4.0

- Research focus: mechanistic interpretability, sparse autoencoders, weight decomposition, feature circuit analysis, faithful reasoning and deception detection, feature geometry.
- Submitted 3 papers to top venues (NeurIPS, EMNLP) within the first semester of the PhD program.

BSc, Computer Science (Talent Program) – Hanoi University of Science and Technology (HUST) Sep 2021 – Sep 2025

GPA: 3.86 / 4.0 · Graduated 2025

- Published 1 first-author paper at ICLR 2025 as an undergraduate.
- Competitive CS Talent Program; with national Olympiad awards in linear algebra and calculus.

PUBLICATIONS & PREPRINTS

[1] (ICLR 2025) [NeurFlow: Interpreting Neural Networks through Critical Neuron Groups and Functional Interactions](#) —

Tue M. Cao, Nhat H. Xuan, Le P. Nguyen, Hieu H. Pham, My T. Thai

↳ Identifies functionally coupled **neuron groups** as interpretable circuit units, traces their functional interactions, and maps how these groups collectively determine model decisions.

[2] (Under review — NeurIPS 2026) [Semantic Optimal Transport for Sparse Autoencoder Feature Matching and Circuit Compression](#) — Tue M. Cao, Nguyen Do, My T. Thai

↳ Matches semantically similar features distributed across all layers via optimal transport, enabling **supernode clustering in circuit tracing (Anthropic's Neuropedia)**. Tracks how features evolve across layers and identifies modular groups with shared functional roles (e.g., **induction features**, bracket detection).

[3] (Under review — EMNLP 2026) [Tree SAE: Learning Hierarchical Feature Structures in Sparse Autoencoders](#) — Tue M. Cao, Nhat H. Xuan, Raed Alharbi, Le P. Nguyen, My T. Thai

↳ Identifies **fundamental limitations** of existing **hierarchical feature definitions** in existing sparse autoencoders. Proposes **tree-structured SAEs** that learn genuinely hierarchical features at multiple granularities, advancing understanding of hierarchical concept geometry in LLMs.

[4] (Under review — NeurIPS 2026) [ReSS: Learning Reasoning Models for Tabular Data Prediction via Symbolic Scaffold](#)

— Chenlang Yi, Gang Li, Zizhan Xiong, Tue M. Cao, Yanmin Gong, My T. Thai, Tianbao Yang

↳ **Faithful and explainable reasoning** for tabular data prediction, guided by decision trees. Results demonstrate that the generated reasoning traces are causally important, non-redundant, and less prone to hallucination.

RESEARCH EXPERIENCE

Research Team Lead — Decomposing Weights to Trace Hallucination Source in Vision-Language Models *Sep 2025 – Present*

Advisor: Assoc. Prof. Le P. Nguyen (HUST)

- Leading an undergrad research team: [Quoc Bao](#) (HUST), [Cong Pham](#) (Nanyang Technological University – NTU).
- Proposed the project direction and methodology for extracting prior knowledge from weight space; designed the main experiments.
- Project details:
 - *Decomposing weights* to identify low-rank regions in weight space that encode prior knowledge driving object hallucination in vision-language models.
 - Validates prior knowledge extraction via activation and attribution patching, attributing hallucinated outputs to the specific weight regions identified.
 - Shows that prior knowledge editing methods (ROME, MEMIT, Nullu) fail to localize prior knowledge, causing language collapse; finetuning-based hallucination methods introduce new mechanisms rather than removing the source.

Research Team Lead — Model Steering Benchmark

Sep 2025 – Present

Advisor: Assoc. Prof. Le P. Nguyen (HUST)

- Leading an undergrad research team: [Phong Quan](#) (HUST), [Hop Le](#) (HUST).
- Proposed the project direction, a flow-matching-based steering improvement, and the evaluation metrics.
- Project Details:
 - Constructs a systematic benchmark for evaluating activation steering methods (*weight steering, flow matching steering, representation finetuning, steering vectors, SAE-based steering*).
 - Existing literature covers 30+ steering methods with no cross-evaluation; prior benchmarks target only simple tasks. This benchmark evaluates a comprehensive set of methods against alignment-relevant tasks including *deception steering, refusal steering, and hallucination steering*.

Main Researcher — Feature Circuit Compression & Feature Matching with Semantic Optimal Transport *Jan 2026 – April 2026*

Advisor: Prof. My T. Thai (UF)

- Proposed the project direction, developed the semantic optimal transport matching framework, and designed the main experiments.
- [Under review in NeurIPS 2026.](#)
- Project details:
 - *Automatically compresses* feature circuits into supernode graphs (as in Anthropic’s circuit tracing) and matches semantically similar features across all layers via optimal transport.
 - Discovers modular feature groups sharing functional roles, including *induction feature groups* and bracket-detection groups scattered across layers.
 - Provides theoretical guarantees for semantic Wasserstein score stability under noise, matching robustness, and optimality of matching within a bounded margin.

Research Assistant — Explain and Improve Autonomous Driving Model

Jan 2026 – April 2026

Advisor: Prof. My T. Thai (UF)

- Explained failure modes of the autonomous driving model and designed experiments to verify the explanations.
- Preparing to submit.
- Project details:
 - Identifies failure modes of a recent autonomous driving model. Key finding: *indecisive in urgent situation*.

- Develops an automatic improvement framework that prompts a VLM to analyze failure explanations and *generate targeted test scenarios* for retraining.

Research Assistant — Faithful and Explainable Reasoning Model for Tabular Data

Jan 2026 – April 2026

Advisor: Prof. My T. Thai (UF)

- Designed evaluation metrics for faithfulness and causal sufficiency of the reasoning trace.
- [Under review in NeurIPS 2026.](#)
- Project details:
 - Uses decision trees and LLMs to *generate faithful, explainable reasoning traces* for tabular data prediction; fine-tunes a specialized model on the generated traces.
 - Evaluation shows that the fine-tuned model improves on faithfulness and explainability metrics while reducing hallucination rate.

Main Researcher — Tree Structured Sparse Autoencoders

July 2025 – Jan 2026

Advisors: Prof. My T. Thai (UF) · Assoc. Prof. Le P. Nguyen (HUST)

- Led all research phases: problem formulation, model design, implementation, and evaluation.
- [Under review in EMNLP 2026.](#)
- Project details:
 - Demonstrates the limitations of existing hierarchical feature definitions and hierarchical sparse autoencoders, motivating a principled redesign.
 - Proposes the first *tree-structured sparse autoencoder* that learns genuinely hierarchical feature pairs while maintaining equal or better performance on key benchmarks, and demonstrates that the structure promotes hierarchical feature geometry.

Main Researcher — NeurFlow: Neuron-Level Circuit Analysis

Jan 2024 – Jan 2025

Advisors: Prof. My T. Thai (UF) · Assoc. Prof. Le P. Nguyen (HUST) · Dr. Hieu H. Pham (VinUni)

- Led all research phases: problem formulation, method design, implementation, and peer-review revision.
- [Accepted at ICLR 2025.](#)
- Project details:
 - Identifies polysemanticity and low per-neuron contribution as fundamental limitations of neuron-level interpretability, motivating group-level analysis.
 - Proposes understanding model computation through *neuron groups* sharing a functional role, and traces group-level circuits to map how decisions propagate through the model.

TECHNICAL SKILLS

Mech. Interp.	TransformerLens · SAELens · causal tracing · attribution graphs · feature circuits · sparse autoencoders
Deep Learning	PyTorch · PyTorch Lightning · HuggingFace Transformers
Research	Experiment tracking (Weights & Biases) · reproducible ML pipelines · scientific writing · LaTeX

HONORS & AWARDS

Strong national-level Olympiad results:

- 2nd place, National Mathematical Olympiad for University Students — Linear Algebra (Vietnam Mathematical Society, 2022)

- 2nd place, National Mathematical Olympiad for University Students — Calculus (Vietnam Mathematical Society, 2022)
- National Team Selection participant, International Physics Olympiad (Ministry of Education, Vietnam, 2021)
- 2nd place, National Physics Olympiad (Ministry of Education, Vietnam, 2021)
- 3rd place, National Physics Olympiad (Ministry of Education, Vietnam, 2020)
- 1st place, Coastal Provinces & Northern Vietnam High School Competition in Physics (Ministry of Education, Vietnam, 2019)

MENTORING

- [Cong Pham](#) (undergraduate – Nanyang Technological University)
- [Quoc Bao](#) (undergraduate – Hanoi University of Science and Technology)
- [Phong Quan](#) (undergraduate – Hanoi University of Science and Technology)
- [Hop Le](#) (undergraduate – Hanoi University of Science and Technology)
- [John Ronchet](#) (undergraduate – University of Florida)
- [Dat Le](#) (undergraduate – University of Florida)
- [Vinh Nguyen](#) (undergraduate – University of Florida)

REFERENCES

Prof. My T. Thai — UF Research Foundation Professor, CISE, University of Florida · mythai@cise.ufl.edu

Assoc. Prof. Phi Le Nguyen — Managing Director, International AI Research Center, HUST · lenp@soict.hust.edu.vn

Dr. Huy Hieu Pham — Associate Director, VinUni-Illinois Smart Healthcare Center · hieu.ph@vinuni.edu.vn